

Probability and Statistics – for Computer Science

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Applied Statistics and Probability for Computer Scientists

Samples and Populations

Chapter 6 — The Sample Mean and Confidence Intervals

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Chapter Outline | What we will cover

Populations & the Sample Mean
Variance of the Sample Mean
Confidence Intervals: The Big Idea

Estimating the Standard Error
The CLT and the Bootstrap
Applications
Summary

Learning Objectives

- ▶ Distinguish a **population** from a **sample**
- ▶ Show the sample mean is **unbiased** and quantify its **standard error**
- ▶ Construct **confidence intervals** with known and unknown variance
- ▶ Apply the **bootstrap** and see CIs used in five real domains

Section 1

Populations & the Sample Mean

Definitions

- ▶ **Population**: the entire set we want to know about
- ▶ **Sample**: the (smaller) subset we actually observe

The urn model

Population = an urn of numbered balls. A sample is drawn (with or without replacement) to estimate the urn's true mean.

Definition

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

Unbiasedness

$$E[\bar{X}] = \mu$$

On average across many samples, $\bar{X}$ hits the true population mean μ exactly.

Section 2

Variance of the Sample Mean

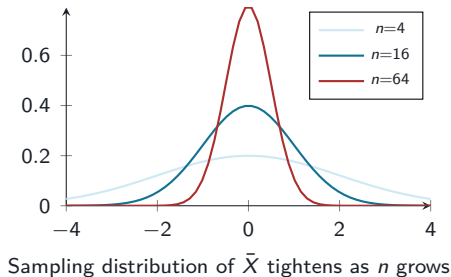
Variance of the Sample Mean | Why bigger samples are better

Key result

$$\text{var}(\bar{X}) = \frac{\sigma^2}{n}$$

Standard error

$$\text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$



Sampling Without Replacement | The finite population correction

Finite population correction

For a population of size N , sample size n :

$$\text{var}(\bar{X}) = \frac{\sigma^2}{n} \cdot \frac{N - n}{N - 1}$$

Rule of thumb

If $n/N < 0.05$ (sample is a small fraction of the population), the correction is negligible — sampling with or without replacement behave the same.

Section 3

Confidence Intervals: The Big Idea

The Big Idea | A range of plausible values, not a single guess

General form

estimate $\pm$ margin of error

$$\bar{X} \pm z^* \cdot SE(\bar{X})$$

Common confidence levels

Level	z^*
90%	1.645
95%	1.960
99%	2.576

Correct interpretation

"95% confidence" means: if we repeated the sampling many times, about 95% of the resulting intervals would contain μ — *not* that this particular interval has a 95% chance of containing it.

Section 4

Estimating the Standard Error

When σ Is Unknown | Use the sample standard deviation

The problem

We rarely know the true population σ — we must estimate the standard error from the data itself.

The solution

Replace σ with the sample std. dev. s :

$$\widehat{SE} = \frac{s}{\sqrt{n}}$$

Known vs. Unknown Variance | z for known σ , t for unknown σ

Case	Confidence interval
σ^2 known	$\bar{X} \pm z^* \frac{\sigma}{\sqrt{n}}$
σ^2 unknown (most common)	$\bar{X} \pm t_{n-1}^* \frac{s}{\sqrt{n}}$

The t -distribution

Wider tails than the normal to account for the extra uncertainty in estimating σ from a small sample; converges to $N(0, 1)$ as n grows.

Section 5

The CLT and the Bootstrap

Why Confidence Intervals Work | The Central Limit Theorem, again

The CLT guarantees $\bar{X}$ is approximately normal for *large enough* n — regardless of the population's own shape. This is what justifies using z^* or t^* multipliers.

How large is "large enough"?

Rule of thumb: $n \geq 30$ is usually safe unless the population is heavily skewed or has extreme outliers.

The Bootstrap Method | Estimating standard error by resampling

The idea

1. Resample n points *with replacement* from the data
2. Recompute the statistic (e.g. the mean)
3. Repeat thousands of times

Why it's useful

Works even when no clean formula for the standard error exists (e.g. for the median or a correlation coefficient).



Section 6

Applications

Five Domains, One Idea | Confidence intervals quantify uncertainty everywhere

Domain	What is estimated
A/B testing (web)	Difference in conversion rate between variants
Quality control (manufacturing)	Mean defect rate / product dimension
Survey sampling (polling)	Population proportion supporting a candidate
Medical trials	Difference in outcome between drug and placebo
ML model evaluation	True model accuracy from a held-out test set

Worked Example: Election Polling | The margin of error

Poll of $n = 1000$ voters: $\hat{p} = 0.52$ support candidate A.

$$SE = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} \approx 0.0158$$

95% CI

$$0.52 \pm 1.96(0.0158) \approx [0.489, 0.551]$$

The race may be a statistical tie — the interval crosses 0.50.

What shrinks the interval?

Larger n , lower confidence level, or smaller σ — width shrinks as $1/\sqrt{n}$, so *quadrupling* n only halves the margin.

Common mistakes

- ▶ Treating the CI as "95% probability μ is inside"
- ▶ Ignoring non-random sampling bias
- ▶ Using z^* with small n and unknown σ



Section 7

Summary

Key Formulas | The Chapter 6 toolkit

Quantity	Formula
Sample mean	$\bar{X} = \frac{1}{n} \sum_i X_i$, unbiased: $E[\bar{X}] = \mu$
Standard error	$SE = \sigma/\sqrt{n}$ (or $s/\sqrt{n}$ if σ unknown)
CI (σ known)	$\bar{X} \pm z^* \sigma/\sqrt{n}$
CI (σ unknown)	$\bar{X} \pm t_{n-1}^* s/\sqrt{n}$

Chapter Summary | Key takeaways

Estimate

The sample mean is unbiased; its spread shrinks as $1/\sqrt{n}$.

Quantify uncertainty

Confidence intervals turn a point estimate into a range with a known long-run coverage rate.

Don't confuse

"95% CI" $\neq$ "95% probability μ is in this interval" — confidence is about the *procedure*.

Looking ahead

Ch. 7: turning intervals into formal **hypothesis tests** and p -values.

Thank you!

Questions & Discussion

Next: Chapter 7 — The Significance of Evidence